PARISHRAM 2.0

CLASS 12TH 2026

Biology-Sexual Reproduction in Flowering Plants

Botany

LECTURE-4

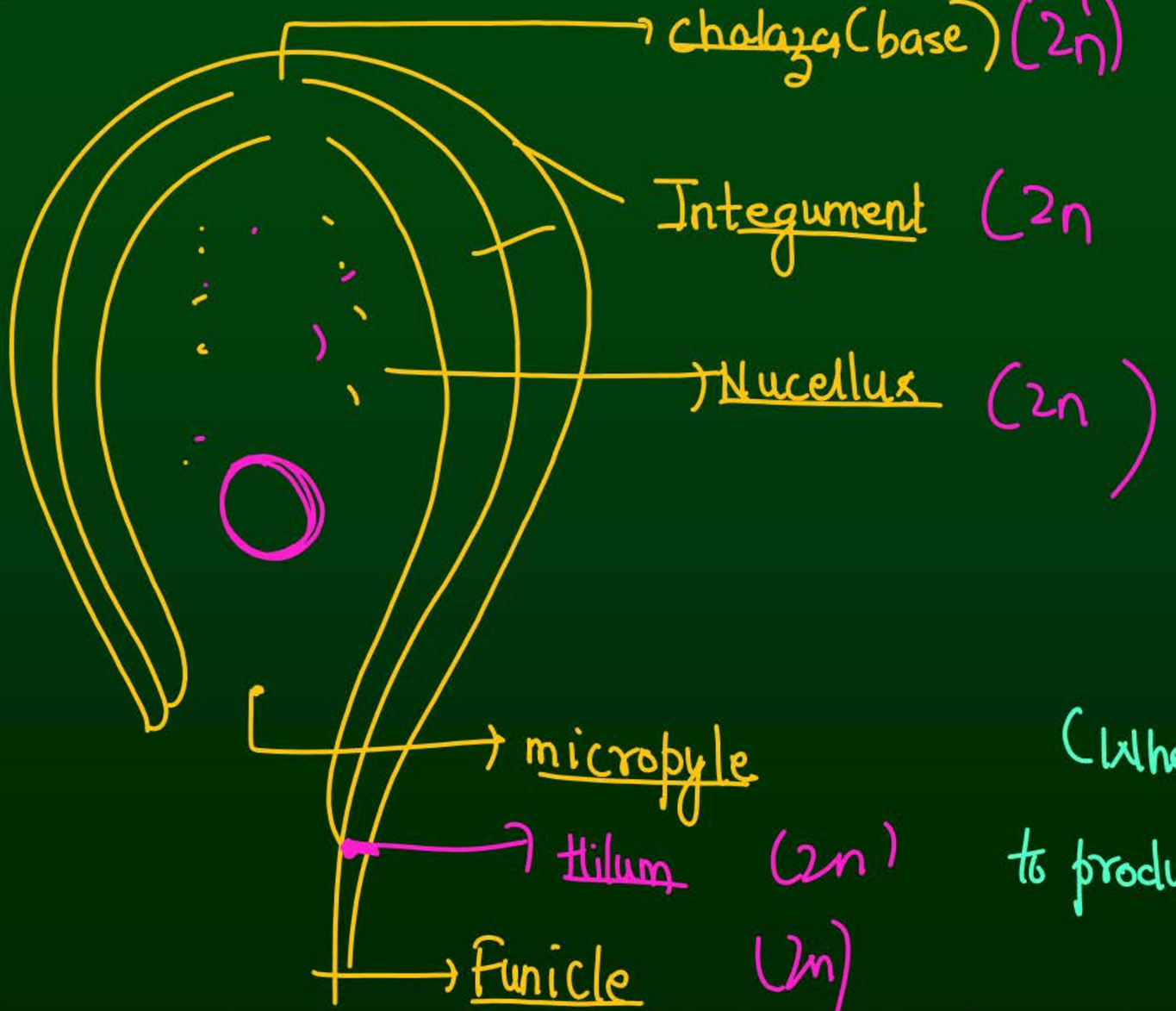
BY-AYUSHI AGGARWAL MA'AM

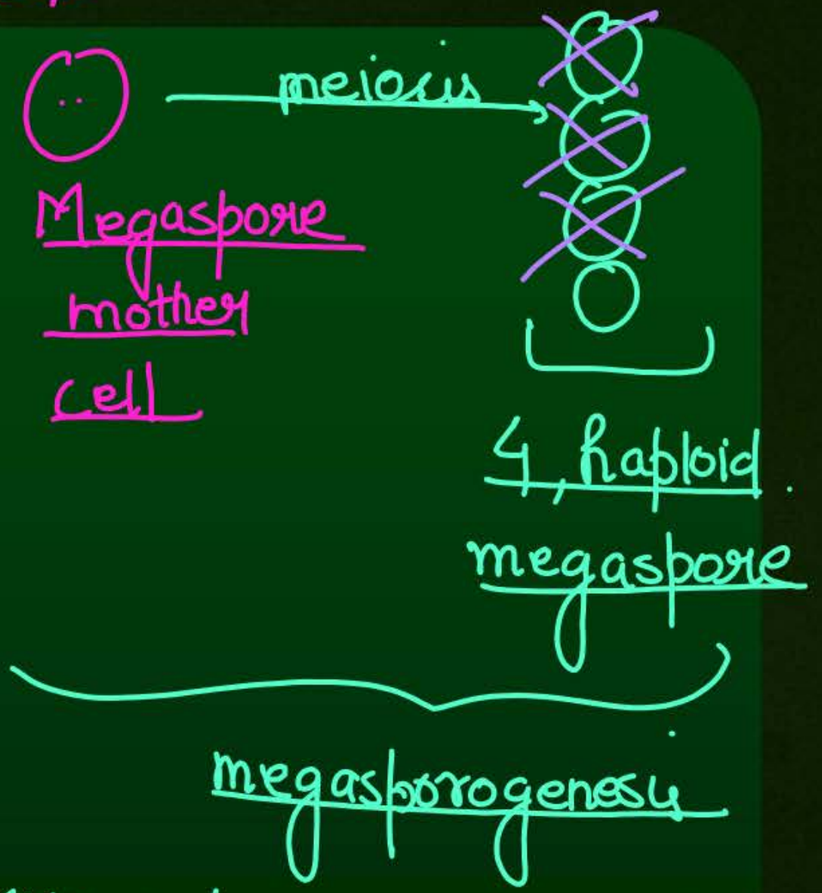


TOPICS TO BE COVERED

1. MEGASPOROGENESIS
2. POLLINATION
3.
4.

MEGASPOROGENESIS





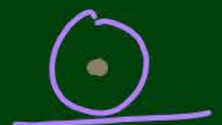
(When MMC undergoes reduction division to produce 4, haploid megaspore)

Out of 4 megaspore, 3 gets degenerated & only one remains functional.

Single megaspore (n) —— mitosis ——→ Embryosac / female gametophyte

Monosporic development.

Monosporic develop.



n, megaspore

3 round of free nuclear mitosis

(repeated karyokinesis & cytokinesis at last)

Embryosac

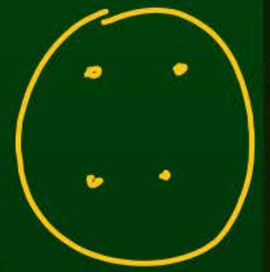


MI → Karyokinesis I



(Binucleate)

MII → Karyo II



Tetranucleat

KIII



(octanucleate)



cytokinesis

(i) Out of 8 nucleus 3 nucleus undergoes cytokinesis to form 3-celled



Chalazal end

Antipodal cell

polar nuclei

central cell

female gamete/egg

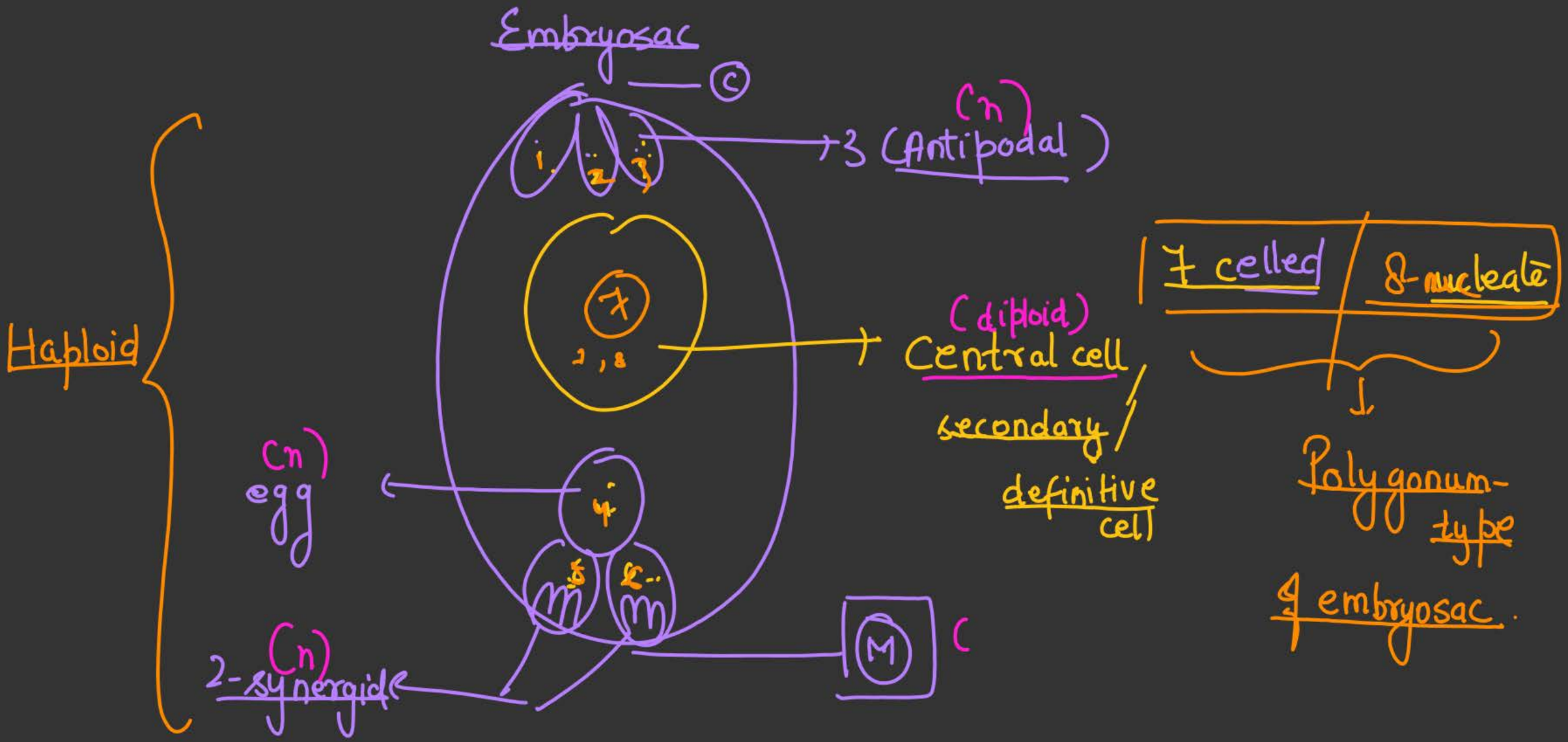
synergid.

Micropylar end

Embryo sac

filiform apparatus (finger-like)

(ii) Again 3 more nucleus undergoes cytokinesis to form 3-celled called egg apparatus at micropylar end


Embryosac
C
(n)
+3 (Antipodal)
Haploid
7
1, 8
(diploid)
Central cell
secondary
definitive
cell
7 celled
8 nucleate
Polygonum-
type
embryosac.
(n)
egg
(n)
2-synergids
M
C

Pollination.

Class XI. Arising from the placenta are the **megasporangia,** commonly called **ovules.** The number of ovules in an ovary may be one (wheat, paddy, mango) to many (papaya, water melon, orchids).

The Megasporangium (Ovule) : Let us familiarise ourselves with the structure of a typical angiosperm ovule (Figure 1.7d). The ovule is a small structure attached to the placenta by means of a stalk called **funicle**. The body of the ovule fuses with funicle in the region called **hilum.** Thus, hilum represents the junction between ovule and funicle. Each ovule has one or two protective envelopes called **integuments.** Integuments encircle the nucellus except at the tip where a small opening called the **micropyle** is organised. Opposite the micropylar end, is the **chalaza**, representing the basal part of the ovule.

Enclosed within the integuments is a mass of cells called the **nucellus**. Cells of the nucellus have abundant reserve food materials. Located in the nucellus is the **embryo sac** or **female gametophyte**. An ovule generally has a single embryo sac formed from a megaspore.

Megasporogenesis : The process of formation of megaspores from the **megaspore mother cell** is called **megasporogenesis**. Ovules generally differentiate a single megaspore mother cell (MMC) in the micropylar region

of the nucellus. It is a large cell containing dense cytoplasm and a prominent nucleus. The MMC undergoes meiotic division. *What is the importance of the MMC undergoing meiosis?* Meiosis results in the production of four **megaspores** (Figure 1.8a).

Female gametophyte : In a majority of flowering plants, one of the megaspores is **functional** while the other three degenerate. Only the **functional megaspore** develops into the **female gametophyte** (**embryo sac).** This method of embryo sac formation from a single megaspore is termed **monosporic** development. *What will be the ploidy of the cells of the nucellus, MMC, the functional megaspore and female gametophyte?*

Let us study about the formation of the embryo sac in detail. (Figure 1.8b). The nucleus of the functional megaspore divides mitotically to form two nuclei which move to the opposite poles, forming the **2-nucleate** embryo sac. Two more sequential mitotic nuclear divisions result in the formation of the **4-nucleate** and later the **8-nucleate** stages of the embryo sac. It is of interest to note that these mitotic divisions are strictly free nuclear, that is, nuclear divisions are not followed immediately by cell wall formation. After the 8-nucleate stage, cell walls are laid down leading to the organisation of the typical **female gametophyte** or **embryo sac**. Observe the distribution of cells inside the embryo sac (Figure 1.8b, c). Six of the eight nuclei are surrounded by cell walls and organised into cells; the remaining two nuclei, called polar nuclei are situated below the egg apparatus in the large **central cell**.

There is a characteristic distribution of the cells within the embryo sac. Three cells are grouped together at the micropylar end and constitute the **egg apparatus**. The egg apparatus, in turn, consists of two **synergids** and one **egg cell**. The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which play an important role in guiding the pollen tubes into the synergid. Three cells are at the chalazal end and are called the **antipodals**. The large central cell, as mentioned earlier, has two polar nuclei. Thus, a typical angiosperm embryo sac, at maturity, though **8-nucleate** is **7-celled**.

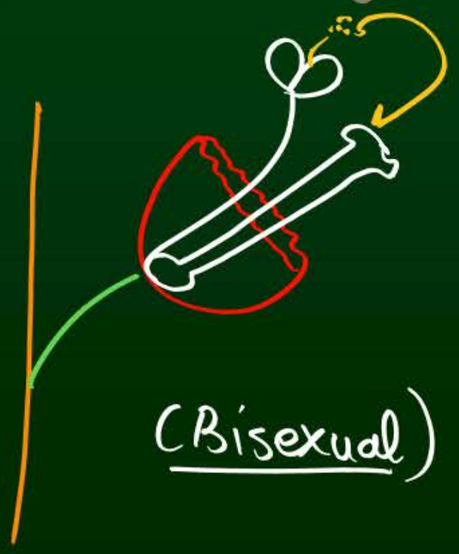
POLLINATION:

Transfer of pollengrains from the anther to the stigma

Same flower

(self pollination / Inbreeding / Autogamy

→ No ext agent is required



(Bisexual)

Diff flower (unisexual)

Geitonogamy

(staminate)

(pistillate)

(Monoecious)

Cross / Allogamy / Out-breeding (Dioecious)

Xenogamy



(a)

(b)

Geitonogamy → is a type of cross pollination where both the flower (♂ + ♀) are born on the same plant

eg Maize / coconut

Xenogamy :- Type of cross pollination where both the flowers are present on diff plant of same species

↓

Dioecious

eg Papaya, datepalm

Which is better

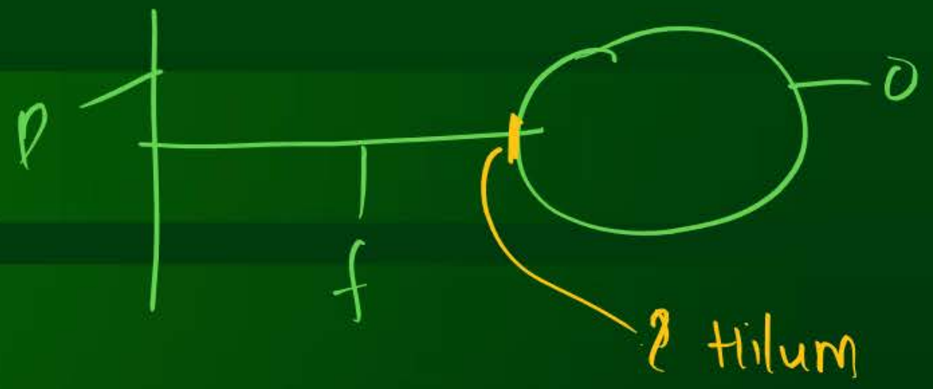
Auto — ?
Cross — ?

Cross pollination (Xenogamy) is better than self because it avoids inbreeding depression as both the gamates are genetically different

QUESTION

The point of attachment of funiculus to the body of ovule is

1. Placenta
2. Micropyle
3. Integument
4. Hilum





QUESTION

The gynoecium *of Michelia* is

1. Monocarpellary
2. Multicarpellary, apocarpous
3. Multicarpellary, syncarpous
4. Bicarpellary, syncarpous



QUESTION

Ovule found in 82% of angiosperm families is

(Most common type of ovule)

1. Anatropous
2. Orthotropous
3. Amphitropous
4. Circinotropous



THANK YOU